HARDWARE PLACEMENT GUIDE

Infidex 176 V x IDENTIDEM.design Remix

Where every heat-set insert, screw, ball plunger, magnet and bonded part goes.

Camera: Infidex 176 V, an open source project by Denis Aminev (Time to Waste). Remix by IDENTIDEM.design.

Every location and screw length below was measured from the remix CAD assembly. Reference codes match the drawings. The remixed body comes two ways, in one piece or as two glued halves, and the hardware is the same for both. Positions are in the CAD assembly frame in millimetres: X runs front (lens) to back (door), Y runs from the take-up end (0) towards the rewind end (−178), Z is up.

I

10 × M2×4 · 4 × M3×4

Heat-set inserts, I1–I14

S

14 screws

6 × M2×6 CSK · 2 × M2×4 CSK · 4 × M3×4 cap or CSK · 1 × M2×6 cap or CSK · 1 × M2×4 thumbscrew

P

6 × D2×3, 7 with the drag cover

Flanged ball plungers, P1–P6, plus P7 on drag-cover

M

1 × Ø5×2 · 2 × Ø4×2

Magnets, M1–M3 · optional, highly recommended

C

Split collar

Take-up spool retainer, C1 · printed part, takeup-washer-stopper

G

Pressure plate

Glued into the door, G1

T

3/8" to 1/4" bush

Tripod adapter, T1 · the socket is printed 3/8"-16

O

Seals and wire

Flocked paper, foam, 0.6–0.8 mm stainless wire · from the original build

Fitting order

1. **All heat-set inserts first** (I1–I11 in the body, I12 in the counter gear, I13 in the inner-ratchet, I14 in the sprocket), before anything else is fitted to those parts. The soldering iron can't then hurt a magnet or loosen a plunger.
2. **Ball plungers** once the body and cover have cooled: P1–P3 in the body's rewind bore, P4–P6 in the cover's take-up bore, and P7 in the tower beside the dial if you printed drag-cover.
3. **Magnets** (optional, highly recommended), checking polarity before gluing. M1 goes into the inner-ratchet from below, after I13 is in.
4. **Pressure plate into the door.** Let the glue cure before closing the door on it.
5. **Take-up spool and printed collar** (takeup-washer-stopper in the files): hold the split collar inside the body under the take-up bore, then lower the spool from the top through it until the collar clicks into the groove.
6. **Two-piece body only:** glue the halves together once their inserts, plungers and magnets are in.
7. **Counter gears** onto their pins in the gear bay, before the top cover goes on.
8. **Screws:** top cover, cold shoe, lens standard, advance knob, then the counter face. With the cover screwed down, turn the sprocket by hand and check the counter train runs freely.
9. **Flocking and foam last,** after a dry run with the back open. The tripod bushing can go in whenever.

I Heat-set inserts

Eleven inserts go into the body: seven M2 in the top face (five for the top cover, two for the cold shoe) and four M3 in the front face for the lens standard. Three more M2 inserts go into moving parts: I12 in the counter gear hub, I13 in the top of the inner-ratchet and I14 in the top of the sprocket. Use 4 mm long inserts everywhere. Every hole is 4.0 mm deep. Buy: Amazon ↗ (M2×4), Amazon ↗ (M3×4, not the common 5.7 mm length).

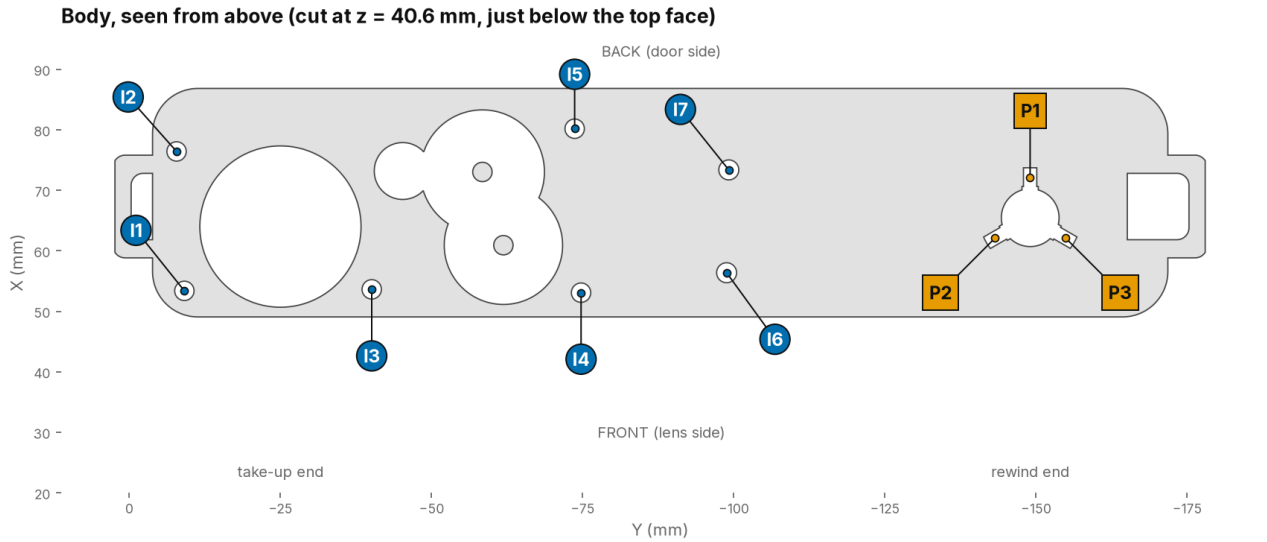


Figure 1 · Body seen from above, lens side at the bottom. I1–I7 are pressed in from the top face.

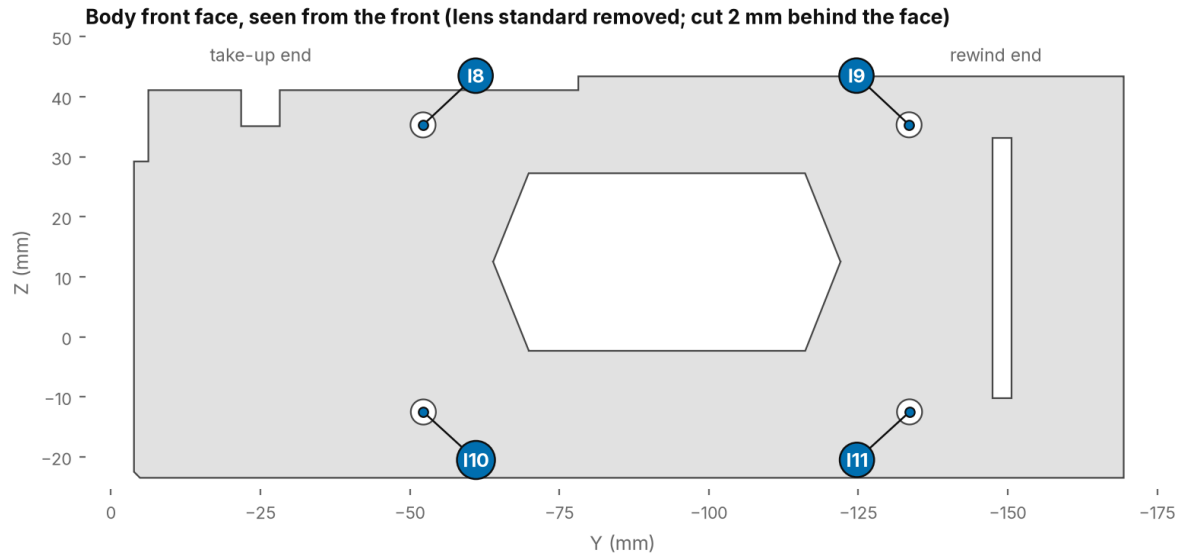


Figure 2 · Front face seen from the lens side, lens standard removed. I8–I11 are pressed in from the front, level with the face.

REF	HOLDS	WHERE	HOLE Ø × DEPTH	INSERT	X / Y / Z
I1	Top cover	Top face, take-up end, front corner	3.2 × 4.0	M2	53.4 / −9.1 / 41.1
I2	Top cover	Top face, take-up end, back corner	3.2 × 4.0	M2	76.4 / −7.8 / 41.1
I3	Top cover	Top face, front edge beside the take-up bore	3.2 × 4.0	M2	53.6 / −40.1 / 41.1
I4	Top cover	Top face, front edge past the gear bay	3.2 × 4.0	M2	53.1 / −74.7 / 41.1
I5	Top cover	Top face, back edge past the gear bay	3.2 × 4.0	M2	80.2 / −73.7 / 41.1
I6	Cold shoe	Top face, centre of camera, front	3.3 × 4.0	M2	56.4 / −98.8 / 42.1
I7	Cold shoe	Top face, centre of camera, back	3.3 × 4.0	M2	73.4 / −99.2 / 42.1
I8	Lens standard	Front face, upper, take-up side	4.2 × 4.0	M3	49.1 / −52.2 / 35.3
I9	Lens standard	Front face, upper, rewind side	4.2 × 4.0	M3	49.1 / −133.5 / 35.3
I10	Lens standard	Front face, lower, take-up side	4.2 × 4.0	M3	49.1 / −52.3 / −12.5
I11	Lens standard	Front face, lower, rewind side	4.2 × 4.0	M3	49.1 / −133.6 / −12.5
I12	Counter face	Counter gear hub, top	3.2 × 4.0	M2	67.7 / −63.9 / 48.0
I13	Advance knob	Inner-ratchet, top hole	3.2 × 4.0	M2	64.0 / −25.0 / 44.0
I14	Sprocket gear	Sprocket, top of the shaft	3.2 × 4.0	M2	73.2 / −45.2 / 30.0

Fitting. For PETG, start the insert tip at about 230 °C. Let the insert sink under its own weight with light pressure, keep it square, and stop when it's flush. Face-down on a flat plate while it's still warm is an easy way to square it up. Stand the body on its base for I1–I7 and lay it on its door side for I8–I11.

I12, I13 and I14. I12's pocket is 4.0 mm deep, so press the insert flush with the top of the hub. I14 goes into the top of the sprocket, and the sprocket gear screws down onto it. I13's hole goes right through into the M1 magnet pocket underneath, so press it flush with the top and check from below that no melted plastic has pushed into the magnet pocket.

P Ball plungers

All of them are D2×3 flanged plungers (Ø2 body, Ø2.5 nose flange, 3 mm long overall, Ø1.5 ball standing 0.4 mm proud of the flange, about 5 N) fitted in radial holes. Amazon ↗. Each of P1–P6 opens into a shaft bore and stops short of the outside wall, so P1–P6 go in *from inside the bore*. The flange sits in its seat flush with the bore wall and the ball points at the shaft.

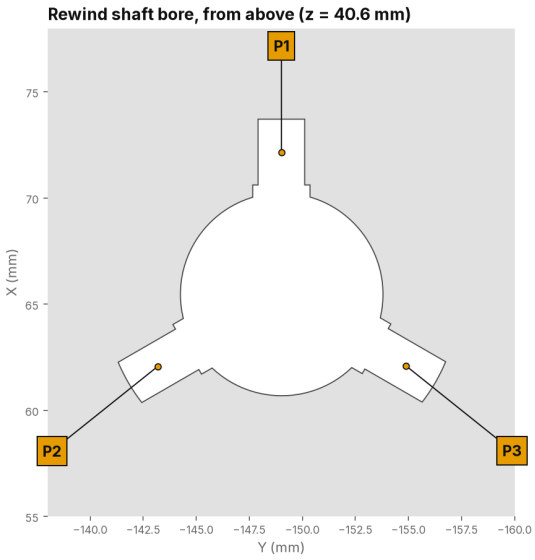


Figure 3 · Rewind bore in the body, 2.8 mm below the top face. Three plungers 120° apart act on the rewind knob.

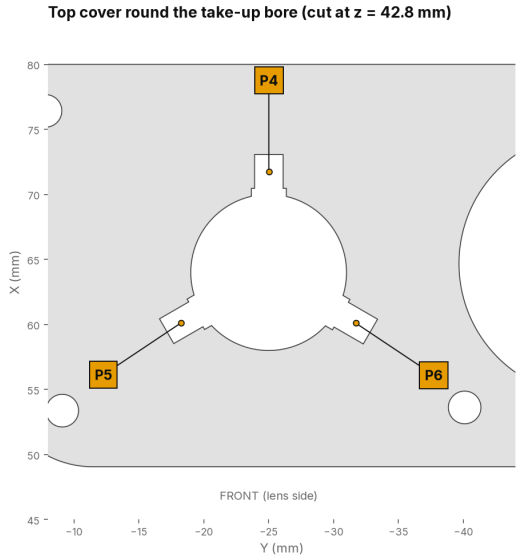


Figure 4 · Take-up bore in the top cover. Three plungers 120° apart, the same pattern as the rewind side, ride in the inner-ratchet's W and R detent grooves.

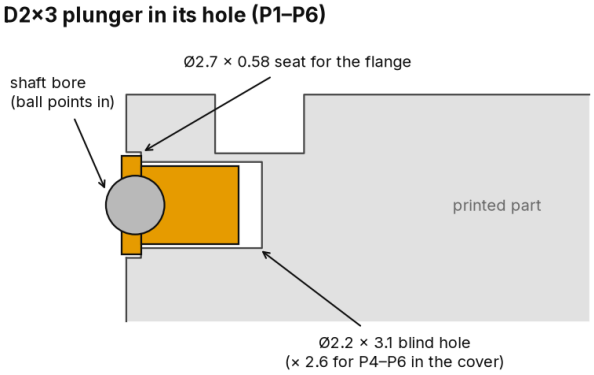


Figure 5 · Cross-section of one plunger in its hole (P1–P6; P7 is shown in Figure 11)

REF	PART	POINTS FROM
P1	Body	Back of the rewind bore
P2	Body	Front, take-up side of the rewind bore
P3	Body	Front, rewind-end side of the rewind bore
P4	Top cover	Back of the take-up bore
P5	Top cover	Front, take-up end side of the bore
P6	Top cover	Front, gear-bay side of the bore
P7	drag-cover only	Tower on the lens side of the dial, at the counter face rim

Fitting. The holes are drawn Ø2.2, 3.1–3.2 mm deep in the body (P1–P3) and 2.6 mm deep in the cover (P4–P6), each behind a Ø2.7 flange seat about 0.6 mm deep. With a 0.4 mm nozzle they print close to Ø2.0, which gives a press fit. Push the plunger body-first from inside the bore with a flat-ended rod until the flange sits in its seat and is level with the bore wall. If one is loose, put a small drop of CA on its body before pressing it in. Keep glue away from the flange and the ball, or the ball will stick. P7's hole is also Ø2.2, 2.8 mm deep behind a Ø2.7 seat set 0.1 mm deeper than the tower face, so its flange stands 0.1 mm proud of the face and 0.1 mm clear of the dial rim. It prints horizontally and may come out slightly oval.

M Magnets Optional, highly recommended

The camera works without them, but they hold the inner-ratchet firmly down on the spool head, so fit them if you can. Buy: Amazon ↗ (Ø5×2), Amazon ↗ (Ø4×2). All three sit on the take-up axis and hold the inner-ratchet down on the take-up spool head. When the camera is assembled, the Ø5×2 magnet faces the Ø4×2 stack across a gap of about 0.3 mm.

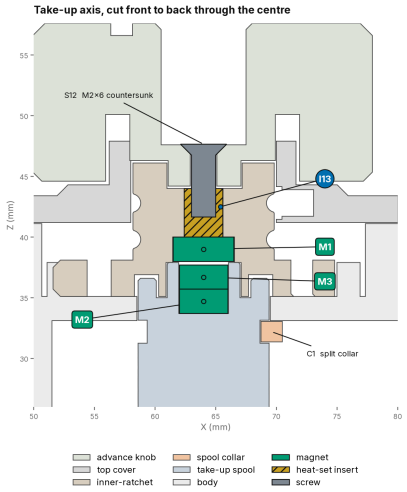


Figure 6 · Take-up axis cut front to back: countersunk knob screw S12 into I13, magnet M1 directly below it, M2 and M3 in the spool head, and collar C1 under the take-up bore.

REF	MAGNET	POCKET	FITS FROM
M1	Ø5 × 2	Ø5.1 × 2.1	Underside of the inner-ratchet, pressed to the pocket floor
M2	Ø4 × 2	Ø4.1 × 4.2	Top of the spool head, at the bottom of the pocket
M3	Ø4 × 2	same pocket	On top of M2, 0.2 mm below the top of the head

Polarity. The top face of the M2/M3 stack has to *attract* the exposed face of M1. Let M2 and M3 snap together into a stack first. Then touch M1 to the top of the stack and mark M1's exposed face with a pen dot before separating them. Glue M1 with the dot facing out of its pocket and the stack with its free face up, then check the attraction with the parts dry-fitted before the glue cures.

S Screws

Each length is the longest standard screw that stays inside its insert without bottoming out. It was worked out from the head seat, the clamped part and the depth of the hole below.

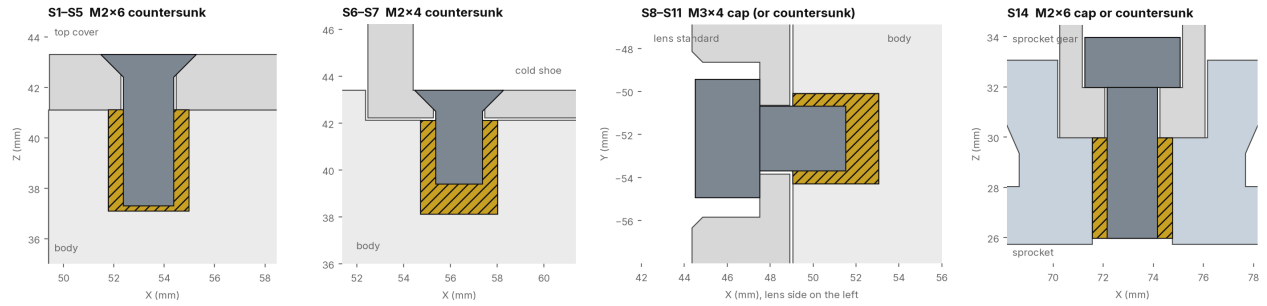


Figure 7 · Screws in their inserts, shown in section. The head seats and hole bottoms are taken from the model.

REF	SCREW	QTY	HOLDS	INTO	ROOM BELOW THE TIP	THREAD IN INSERT	BUY
S1–S5	M2×6 countersunk (ISO 10642)	5	Top cover to body	I1–I5	0.2 mm	3.8 mm	Amazon ↗
S6–S7	M2×4 countersunk (ISO 10642)	2	Cold shoe to body	I6–I7	1.3 mm	2.7 mm	Amazon ↗
S8–S11	M3×4 socket cap (ISO 4762) or countersunk (ISO 10642)	4	Lens standard to body	I8–I11	1.5 mm	2.5 mm	Cap ↗ CSK ↗
S12	M2×6 countersunk (ISO 10642)	1	Advance knob to inner-ratchet	I13	1.65 mm to M1	2.35 mm	Amazon ↗
S13	M2×4 thumbscrew, flat underside	1	Counter face to counter gear	I12	0.9 mm	3.1 mm	Amazon ↗
S14	M2×6 socket cap (ISO 4762) or countersunk (ISO 10642)	1	Sprocket gear to sprocket	I14	0.25 mm	4.0 mm	Cap ↗ CSK ↗

Lengths. The cold shoe (M2×4) and lens standard (M3×4) lengths are proven on the printed camera and leave plenty of clearance. The M2×6 cover and sprocket screws are the longest that fit: the next size up bottoms out. The S1–S5 figures are with the cover screwed down, which seats it 0.1 mm below its drawn height. S14 works as a socket cap or a countersunk head, whichever you have, and either way it takes the full depth of its insert, so run it down gently and stop when it seats. Don't go longer on S12 either: an M2×8 would reach 0.35 mm into the M1 magnet pocket.

Heads. The knob's seat for S12 is a Ø3.7 × 90° countersink 0.75 mm deep, so a standard M2 flat head (Ø3.8) finishes level with the knob face and takes 2.35 mm of thread. The countersinks in the cover and cold shoe are Ø3.7–3.8 at 90°, so M2 flat heads finish flush or a hair proud. The lens-standard counterbores are Ø7.3 × 3.1 deep, so either an M3 cap head (Ø5.5 × 3) or an M3 countersunk head sits below the surface.

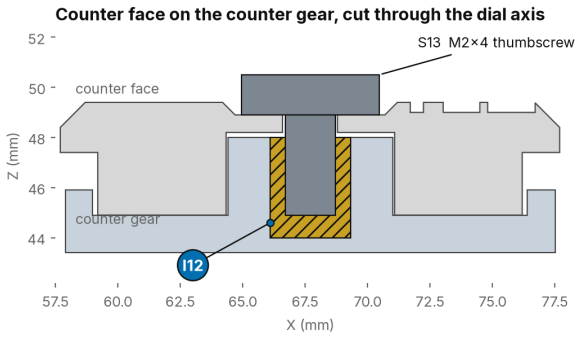


Figure 8 · Counter face held by thumbscrew S13 in insert I12. The head sits in a Ø6.0 × 0.5 mm recess, so a head up to about Ø5.8 sits down in the face.

> **Counter face.** The face starts at 25 and counts up: 25 is the loading position, then winding on takes you to 1, 2, 3 and so on. There is no zero. The 20-frame face, on its own pair of idlers (**counter-coupling-gear-20**, **counter-idler-1-20**), works the same way from 20. Lay the face on the hub with the loading number against the marker and finger-tighten S13: the marker is the triangle recessed into the cover beside the dial, in the same place on both covers. The face grips on friction, so to reset it for a new roll you loosen the thumbscrew, turn the loading number back to the marker, and tighten again. Don't use thread lock.

C Take-up spool collar

Printed part, in the files

C1 is a printed part, not a bought one: print it with the rest of the camera from the files in the repo, where it is named takeup-washer-stopper. It is a split ring (Ø9.5 bore × Ø13.0 × 1.7 mm) that stops the take-up spool lifting out of the body. It sits in the groove just below the spool head (Ø9.3 at the root), directly under the body's take-up bore. The collar is wider than the bore, so once it's in, the spool can't pull up through the top.

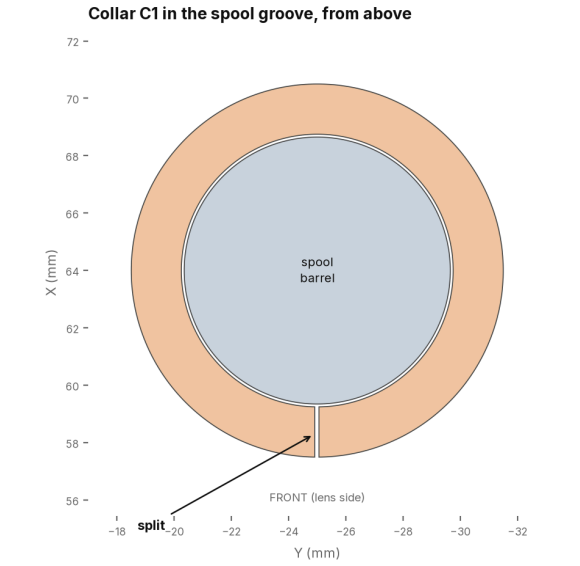


Figure 9 · Collar in the groove, from above. The split is about 0.15 mm wide.

1. Open the body and hold the collar inside it, directly under the take-up bore, with the split facing the lens side.
2. Lower the take-up spool into the body from the top. As it comes down, its bottom end passes through the collar, and the split lets the collar open over the Ø10.8 barrel.
3. Keep lowering the spool until the collar snaps closed in the groove below the head, then spin the spool to check it turns freely without lifting.

G Pressure plate

G1 is one printed part: a rigid platen on a single thin leaf. The leaf joins the platen in the middle and has a bond pad at each end. Glue only the two end pads to the door, never the platen or the middle of the leaf. The door's inside face has a shallow 0.2 mm recess (83.6 × 33.5 mm) that locates the plate, leaving about 1 mm spare at each end and 0.25 mm top and bottom.

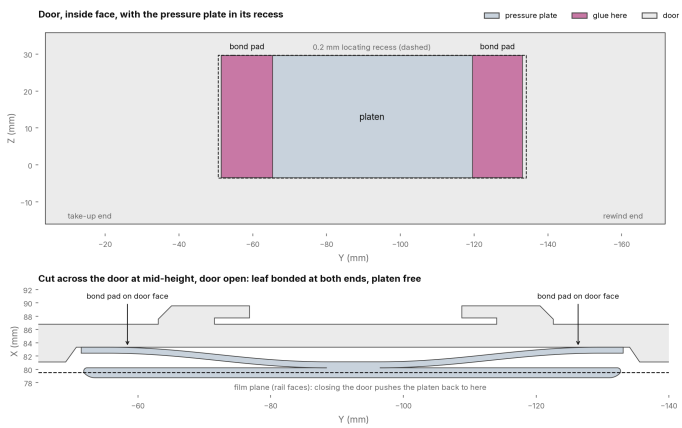


Figure 10 · Pressure plate in the door. Glue goes only on the two end pads, each about 14 × 33 mm, seated on the door face. The lower cut shows the door open, with the platen free.

> **Gluing.** Clean both surfaces with IPA. Put a thin line of CA or a strip of thin double-sided tape on each end pad only. Seat the plate square in the recess with the platen facing the film, and press on the pads alone for 30 seconds. Keep glue out of the gap under the leaf, or the plate stops flexing.

> **Seating and preload.** Both pads sit flat on the door's inside face in the recess. With the door open, the leaf holds the platen 1.6 mm proud of its working position. Closing the door on loaded film pushes the platen back onto the film rails and flexes the leaf by about 0.8 mm, which gives the pressure on the film. Don't shim or glue anything under the leaf to take up that gap.

P7 Counter dial: simple or drag

Choose one pair

The counter comes two ways, and you print one pair: **simple-cover** with **simple-counter-face**, or **drag-cover** with **drag-counter-face**. Everything else is shared: the same body, gears, inserts and screws. Both faces also come in a 20-frame version (**-20**) for the 20-frame idlers, and every face comes as a *-two-colour* file with its marker inlay.

	SIMPLE	DRAG
Extra hardware	None	One D2×3 ball plunger, P7
How the dial moves	Turns only with the gear train	P7's ball presses on the face's plain rim, with a detent flute at every frame
Marker for the loading number	Triangle recessed into the cover	The same triangle, 72° round from P7: five frames on a 25-frame face, four on a 20-frame face
Best for	Fewest parts, lightest wind	A dial that stops in the same place every time and doesn't flutter
Costs	The gear backlash lets the face rock a fraction of a frame	One more plunger, a slightly heavier wind, a tower 1 mm above the face

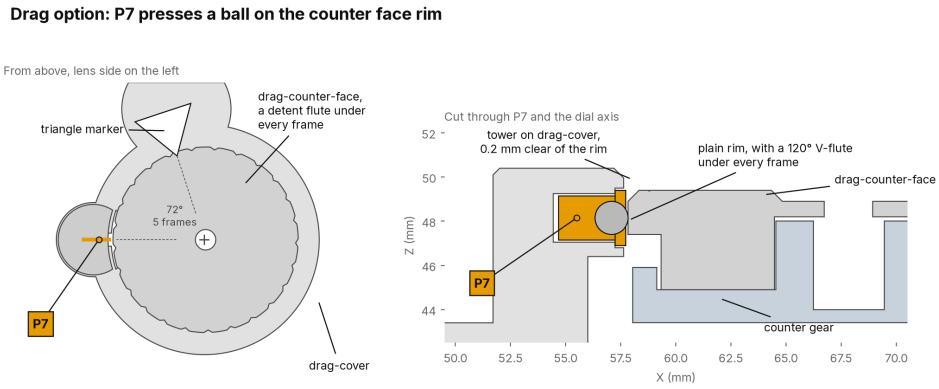


Figure 11 · Drag option, as fitted: P7 pushes the face and the counter gear 0.14 mm across, until the face meets the cover step on the far side. P7 sits in the tower on drag-cover, which then stays 0.2 mm clear of the rim. Its flange stands 0.1 mm proud of the tower face and 0.1 mm clear of the rim, and its ball runs on drag-counter-face's plain rim, dropping 0.15 mm into a 120° V-flute under each frame marker. The triangle marker is exactly five frames (72°) round from P7, so a flute sits under the ball whenever a number sits at the triangle.

> **Fitting the drag option.** Press P7 in from the dial side before the counter face goes on. Then tighten S13 firmly by hand: it has to hold the face against P7's friction and the pull of each flute as well as the gears. Wind through a few frames and check the face stays with the train; if it slips on the hub, tighten S13 a little more. Print the pair together: the simple face has no flutes, so on the drag cover it drags without clicking, and the drag face needs the drag cover's tower, which sits 0.14 mm nearer the dial than the 2.1.0 one.

G2 Two-piece body seam

Two-piece body only

Skip this if you printed the one-piece body. The two halves meet on a flat seam about 34 mm above the base. The bottom half carries a 2 mm ridge that runs right around the film chambers and the film path; the top half has the matching groove. Light arriving along the glue line has to climb over that step to reach the film, which is why only the chambers and the path are ridged and the rest of the seam is plain.

Four alignment pegs set the halves: Ø3.8 × 2 mm posts inside the top half's groove, dropping into Ø4 mm holes in the ridge. Dry-fit before any glue. The halves should sit down with no rock and no light gap at the seam; if they rock, the ridge is not seating and the groove needs a clean-up rather than more clamp.

> **Gluing.** Fit every insert, plunger and magnet first, while the halves are still separate and easy to hold. Then run a thin bead of CA or epoxy on the flat land outside the ridge, press the halves together on the pegs, and clamp lightly until it cures. Keep glue off the ridge crest and out of the groove, or the halves will stand proud and leave the gap you were trying to close.

T Tripod socket

The socket is printed, not an insert: a **3/8"-16 thread** moulded straight into the base on the centreline, five turns and 8.5 mm deep, at 67.4 / −86.1 / −23.5. Nothing is pressed or glued into it.

Most tripods and quick-release plates are 1/4"-20, so fit a **3/8" to 1/4" bushing** and leave it in. Amazon ↗

> **First thread.** Start the bushing by hand and stop as soon as it seats. A printed thread strips if you drive it with a tool, and a 1/4" screw run straight into the 3/8" hole will tear it out. If the thread is tight from the print, run the bushing in and out a couple of times to clear it rather than cutting it.

O From the original build

These are unchanged from Denis's build and are not in the printed files. The remix needs all of them.

ITEM	WHAT FOR	BUY
Black flocked self-adhesive paper	Lining the film chambers and the inside of the door to kill reflections and stray light. Full-size templates are in flocking/ in the repo.	Amazon ↗
Black light-seal foam	Sealing the door where it closes onto the body	Amazon ↗
Rigid stainless wire, 0.6–0.8 mm	Springs and linkages in the original mechanism	Amazon ↗
Superglue (CA)	Magnets, the pressure plate pads, and the body seam on the two-piece body	Amazon ↗

> **Order.** Flocking and foam go in last, after everything else is fitted and the camera has been dry-run with the back open. They are the one thing you cannot fit around.

Infidex 176 V x IDENTIDEM.design Remix. Camera: Infidex 176 V, an open source project by Denis Aminev (Time to Waste). Remix and guide by IDENTIDEM.design.

Buy links open an Amazon UK search for that exact part rather than one listing, so check the size, head type and length before ordering. Measured from the remix CAD assembly, 27 September 2026. Where the CAD and a printed part disagree, go by the print.